Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/31/2006
PI Name:	Nalette, Timothy Ph.D.		
Project Title:	Advanced CO2, Humidity, and Trace Contaminant Control System for Closed Loop Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Windsor Locks	**State:**	CT
Zip Code:	06096-1010	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	08/11/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Papale, William (Hamilton Sundstrand)		
Grant/Contract No.:	NNJ04HF73A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Conduct basic research associated with amine based sorbents for combined humidity and carbon dioxide control for closed loop life support. The basic research will investigate alternate sorbent formulations and configurations that will address enhanced kinetics and loadings to enhance overall system performance. These enhancements will result in reduced system mass, volume and power, while also enhancing overall vehicle integration issues associated with the addition of the humidity control function. Based on the results of the basic sorbent research, we will design, fabricate, assemble and test a full scale, deliverable prototype system.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	May have applicability to CO2 capture from stationary power plants to address greenhouse gas abatement.
Task Progress:	In November 2005, NASA redirected the program to focus on CEV design and the elimination of the trace contaminant component of the original plan. The baseline sorbent SA9T was selected and the hardware design, fabrication, test and delivery of the first unit were accelerated with a desired delivery date of June 2006. The balance of the initial trace contaminant study being conducted by MSFC would be completed, but no additional research associated with trace contaminant control would be pursued. Finally, the universities would continue with basic research associated with the amine chemistry through the second year as originally planned. NASA accelerated the program with the desire to complete the hardware delivery of a complete CO2 and humidity control system by June 2006. In addition to the accelerated hardware and test schedule, the program was also modified to deliver a sub-scale 4 bed test article and a second full scale system sized for CEV. The performance requirements were finalized and the canister design was initiated in Novemeber 2006. The canister fabrication and assembly was completed in May 2006 and testing conducted in May and June 2006. The canister exceeded all performance requirements and was delivered to NASA JSC in mid June, after completing approximately 20 parametric test points. The trace contaminant study conducted at MSFC completed single and multi-component testing of representative contaminants. The preliminary findings were that while the sorbent was able to remove some contaminants with high efficiency, some were only removed approximately 50%, and methane and carbon monoxide were not removed at all. Since the testing involved equilibrium isotherm loadings, and some compounds were not removed at all, it was decided that the amine sorbent would not be considered for trace contaminant control. The balance of the research effort for the second year was concentrated at the University of Connecticut and the University of Hartford. The research efforts were focused on the synthesis of amine sorbents with a high concentration of secondary amine groups and essentially were functionalized sorbents as opposed to the baseline sorbent which is an immobilized liquid in a porous support. The results with both university approaches were disappointing as the relative performance was less than 10% of the baseline sorbent. The effort is currently in the process of being completed, currently planned for early in the 4th quarter of 2006. Finally, numerous design reviews were conducted at NASA JSC for the deliverable hardware, and the PI supported numerous CEV related meetings to act as a consultant for the amine based CO2 and humidity control technology.
Bibliography Type:	Description: (Last Updated: 02/19/2009)
Dissertations and Theses	Arsenault S, Weiss R. "Polymers for CO2 Absorption." Thesis, University of Connecticut, December 2006. , Dec-2006
Papers from Meeting Proceedings	Nalette T, Papale W, Smith F, Perry J. "Development Status of Amine-based, Combined Humidity, CO2 and Trace Contaminant Control System for CEV." 36th International Conference on Environmental Systems, Norfolk, VA., July 17-20, 2006. ICES 2006-01-2192, SAE International, 2006. , Jul-2006